

Research Proposal

Practical Implementation of Machine Learning Models for Adaptive Learning and Intelligent Security

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1. Motivation

With the increasing reliance on digital platforms for education and services, artificial intelligence is playing a central role in shaping user experience and system reliability. My primary objective in applying for this research internship is to gain hands-on experience in designing, training, and deploying machine learning models in real-world systems. I am particularly interested in exploring how ML techniques can be applied both to adaptive STEM learning systems and to intelligent cybersecurity mechanisms.

2. Research Interest and Problem Direction

In the context of STEM education, one practical challenge is building systems that can adapt to individual learning patterns rather than following static progression rules. I am interested in experimenting with predictive models that estimate student mastery levels based on interaction features such as response time, error frequency, and concept-wise performance trends.

The focus would be on implementing supervised learning models to predict learning outcomes and evaluating their performance using appropriate metrics. Further, I would like to explore reinforcement or sequential decision-making approaches for dynamically recommending content based on predicted mastery levels.

In parallel, I am interested in applying machine learning techniques to detect anomalous system behavior in web-based platforms. Modern systems generate large volumes of user interaction data, and understanding how to model normal behavioral patterns is a practical and challenging task. I aim to work on implementing anomaly detection models using behavioral features such as login frequency, session duration, access patterns, and database queries. Techniques such as clustering, isolation-based methods, or neural-network-based models can be evaluated for effectiveness and computational efficiency.

3. Implementation-Oriented Approach

Rather than limiting the work to theoretical modeling, my goal is to understand the complete pipeline:

- Data preprocessing and feature engineering
- Model selection and training
- Hyperparameter tuning and evaluation
- Deployment considerations in a scalable backend system

Through my prior experience developing full-stack applications with authentication and role-based access control, I have gained exposure to backend system design. During this internship, I

hope to bridge that experience with practical machine learning implementation, understanding how models interact with production systems and what trade-offs arise in terms of latency, scalability, and robustness.

4. Expected Learning Outcomes

Through this research internship at IIT Kanpur, I aim to develop a deeper understanding of:

- Practical machine learning experimentation
- Model evaluation and generalization challenges
- Deployment of ML models in real-world systems
- Integrating adaptive intelligence with secure system design

This opportunity would allow me to strengthen my theoretical foundation while gaining meaningful hands-on exposure to intelligent system development under expert mentorship.

5. Conclusion

This proposal reflects my interest in applying machine learning techniques to impactful real-world domains such as education and cybersecurity. I am eager to learn through implementation, experimentation, and system-level integration, and I believe the Department of Intelligent Systems at IIT Kanpur provides the ideal environment to develop these capabilities.